

Component	Metric	Value
Visual (YOLOv8-s)	macro F1@0.5	0.606
	mean AP@0.5	0.519
	mean inference (ms)	35.3
Acoustic (Whisper-Base)	secondary-speaker AUC	0.774
	secondary-speaker F1	0.779
	overlap AUC	0.771
	whisper AUC	0.935
Behaviour (FaceMesh)	off-screen detection AUC	0.938
	off-screen F1	0.873
	lip-movement AUC	0.663
Fusion (random_forest)	F1 (5-fold CV)	0.905 ± 0.041
	ROC AUC	0.886
	PR AUC	0.968
System latency	YOLOv8-s GPU/MPS (ms)	37.7 ± 17.0
	YOLOv8-s CPU (ms)	47.1 ± 4.9
	Whisper buffer (ms)	119.3
	End-to-end FPS (GPU)	23.7
	End-to-end FPS (CPU)	19.4
	Peak RSS (MB)	1102
Privacy	Raw-audio re-ID AUC	0.994
	Metadata re-ID AUC	0.478